CSCE 411: Design and Analysis of Algorithms

Week 13: Approximation Algorithms and Optimization

Date: April 14, 2026 *Nate Veldt, updated by Samson Zhou*

1 Linear programming

A linear program is a mathematical optimization problem with

- A linear objective function
- Linear constraints

We will use x_i to denote variables—unknowns that we need to find to make the objective function as large as possible, and such that the constraints hold.

Examples

Warm-up problem (from CLRS). As a politician seeking approval ratings, you would like the support of 50 urban voters, 100 suburban voters, 25 rural voters.
For each $1 spent advertising one of the following policies, the resulting effects are:

Policy	**Urban**	**Suburban**	**Rural**
Zombie apocalypse	-2	+5	-3
Shark with lasers	+8	+2	-5
Flying cars roads	0	0	+10
Dolphins voting	+10	0	-2

Minimize the amount spent to achieve the desired voter support. How can we write this as an algorithmic or mathematic problem?

Create variables:

- Let _____ be the money spent on ads for preparing for a zombie apocalypse
- Let _____ be the money spent on ads for sharks with lasers
- Let _____ be the money spent on ads for roads for flying cars
- Let _____ be the money spent on ads for allowing dolphins to vote

Then the objective of the problem is:

The constraints of the problem are:

Another example problem. The students of **CSCE 411** are managing their semester project portfolios, which involve three types of projects:

- **Project A:** Basic Algorithms.
- **Project B:** Graph Algorithms.
- **Project C:** Complexity.

Each project type earns a different amount of grade contribution points and consumes a mix of three limited resources: *Self-Study Time*, *Computation Credits*, and *Team Collaboration Hours*.

Project Type	**Grade Points**	**Time (hours)**	**Credits (units)**	**Collab Hours**
A	10	5	3	2
B	12	6	2	4
C	8	4	4	3

The semester budget for a student is:

- Maximum 60 hours of Self-Study Time.
- Maximum 30 Computation Credits.
- Maximum 36 Collaboration Hours.

Additionally, the professor requires that at least 2 projects from each type be completed for a balanced learning experience. Given the budget and these constraints, devise a project plan to maximize the grade.

Write a linear program for this problem.

2 Types of Linear Programs

Linear programs have many variations:

- The objective function can be a maximization or a minimization problem
- The constraints can be equality or inequalities
- Often times, the variables will be greater than or equal to zero

If a particular solution $\overline{x}$ satisfies all of the constraints, we call it a _____.
Otherwise, we call it an _____. The set of solutions that are feasible is called the _____.
Actually, we can perform different conversions to

- Turn a maximization problem into a minimization problem
- Turn an equality constraint into inequality constraints
- Turn an inequality constraint into an equality constraint
- Turn an unconstrained variable into positive variables

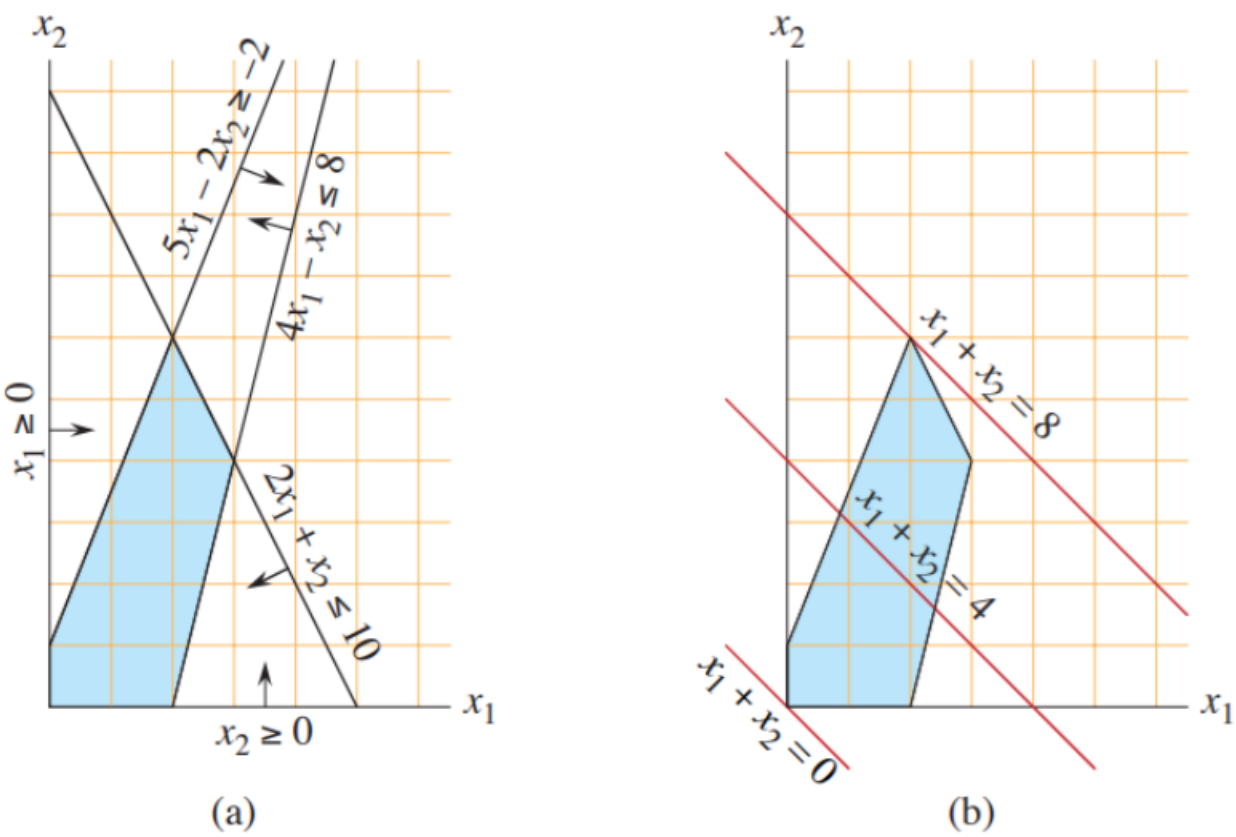
3 Solving a Linear Program

Important Fact: A linear program can be solved in polynomial time, e.g., the ellipsoid algorithm. There are other practical implementations, e.g., the simplex algorithm, that can run in exponential-time in the worst case.

Consider the linear program:

$$\begin{aligned}
\text{Minimize:} \quad & x_1 + x_2 \\
\text{Subject to:} \quad & 4x_1 - x_2 \leq 8 \\
& 2x_1 + x_2 \leq 10 \\
& 5x_1 - 2x_2 \leq -2 \\
& x_1, x_2 \geq 0 \quad \text{(Non-negativity)}
\end{aligned}$$

Intuition: The optimal solution must occur at a _____ of the feasible region.



4 Duality

Question 1. Consider the linear program:

$$\begin{aligned} \text{Maximize:} \quad & x_1 + 2x_2 + 3x_3 + 4x_4 \\ \text{Subject to:} \quad & 10x_1 \leq 10 \\ & x_2 \leq 200 \\ & 5x_1 + 10x_2 + 15x_3 + 20x_4 \leq 15 \\ & x_1, x_2, x_3, x_4 \geq 0 \quad \text{(Non-negativity)} \end{aligned}$$

What can we say about the optimal value?

A 1

B 3

C 10

D 15

E 200

F The limit does not exist

Consider the linear program:

$$\begin{aligned} \text{Maximize:} \quad & 3x_1 + x_2 + 4x_3 \\ \text{Subject to:} \quad & x_1 + x_2 + 3x_3 \leq 30 \\ & 2x_1 + 2x_2 + 5x_3 \leq 24 \\ & 4x_1 + x_2 + x_3 \leq 36 \\ & x_1, x_2, x_3 \geq 0 \quad \text{(Non-negativity)} \end{aligned}$$

What can we say about the optimal value?

What if we take the original constraints and add the first two constraints?

The primal solution must be at most _____

Can we find a linear combination of the equations that exactly matches the objective?
Create variables:

- Let _____ be the _____
- Let _____ be the _____
- Let _____ be the _____

What should be the objective?

The _____ of a linear program is an associated optimization problem where _____ and _____, providing bounds on the _____

In general, for a primal LP

the corresponding dual LP is

Theorem 4.1 (Weak duality)**.** *Let the primal linear program be a minimization problem and its dual be a maximization problem. If x is a feasible solution to the primal and y is a feasible solution to the dual, then*

$$c^\top x \geq b^\top y.$$

Theorem 4.2 (Strong Duality)**.** *If the primal linear program has an optimal solution over a feasible region $\mathcal{P}$ and satisfies the necessary regularity conditions (e.g., feasibility), then the dual also has an optimal solution over the feasible region $\mathcal{D}$, and the optimal values of the primal and dual objectives are equal. That is,*

$$\min_{x \in \mathcal{P}} c^\top x = \max_{y \in \mathcal{D}} b^\top y.$$

5 Graph problems as mathematical optimization problem

We can write the maximum s-t flow problem as the following optimization problem:

The weighted vertex cover problem can be written as the following optimization problem:

Question 2. *Which of the above two problems is a linear program?*

A *The first*

B *The second*

C *Both*

D *Neither*

6 Linear Programming Relaxation for Weighted Vertex Cover

This is the integer program for the weighted vertex cover problem:

Question 3. *Let $\hat{C}$ be the optimal solution value to the linear programming relaxation of the weighted vertex cover integer program, and let C^* be the optimal solution to the weighted vertex cover problem. Which of the following is always true?*

A $\hat{C} \leq C^*$

B $\hat{C} \geq C^*$

C $\hat{C} < C^*$

D $\hat{C} > C^*$

E $\hat{C} = C^*$

7 The Algorithm

WeightedVertexCoverApprox($G = (V, E)$)

1. Solve the linear programming relaxation of the weighted vertex cover problem
2. For each node $v \in V$, if $x_v \geq 1/2$, add v to a node set S. In other words, define

$$S = \{v \in V : x_v \geq 1/2\}$$

3. Return S as a vertex cover

Theorem 7.1. *Let C^* be the weight of the minimum weighted vertex cover of G. The algorithm* WeightedVertexCoverApprox *runs in polynomial time in terms of the size of G and outputs a vertex cover $S \subseteq V$ satisfying $\sum_{v \in S} w_v \leq 2C^*$.*

8 Approximation Algorithms for Optimization Problems

There are so many hard problems out there without known polynomial time solutions! Is all hope lost? What do we do?

Approximation algorithms "quickly" find an answer that is "close" to the solution.

We are now focused on optimization problems, rather than just their *decision* versions.

Definition Let Q be a computational minimization problem (e.g., find a minimum vertex cover, find a minimum s-t cut) and assume that C^* is the optimal (minimum) solution to the problem. An _____ for Q with approximation factor p is an algorithm that

- Runs in
- Outputs a solution value C that is guaranteed to satisfy

Question 4. *What is a lower bound we must assume holds for the value of p when defining an approximation algorithm?*

A *No lower bound is needed, p can be any real number*

B $p \geq -1$

C $p \geq 0$

D $p \geq 1$

E $p \geq C^*$

We can also have approximation algorithms for maximization problems, but in this case C^* represents the *maximum* value (i.e., the solution) for the problem and _____

Wait, is this even possible? We want to guarantee that $\frac{C}{C^*}$ is small in polynomial time, but finding C^* is NP-hard. How can we do that?

To design an approximation algorithm, we need two pieces:

1. _____ A procedure for finding a value $\hat{C}$ that satisfies _____

   - It should take polynomial time

   - This method _____ solve the original problem, but is often related

2. _____ An algorithm for the original problem that returns

   a suboptimal solution _____

   - It should take polynomial time

   - This method often:

We then must prove that _____ and this provides a p-approximation!

Caveat. Often, the lower bounding procedure is *explicit*: there is an actual algorithm that computes the lower bound, and then the upper bounding algorithm explicitly uses the lower bound. However, in some cases no explicit lower bound is computed, and instead one shows implicitly that the upper bounding algorithm has to provide a solution that is better than some lower bound, even though the lower bound isn't computed explicitly. This can be tricky, but it is often possible!

9 Matchings and Vertex Covers

Let $G = (V, E)$ be an undirected and unweighted graph.

A matching $\mathcal{M} \subseteq E$ is a set of edges such that no two edges share the same node.

A vertex cover $\mathcal{C} \subseteq V$ is a set of nodes such that each edge $(u, v) \in E$ has at least one node in $\mathcal{C} \subseteq V$.

Lemma 9.1. *Let $\mathcal{M}$ be a matching of G and $\mathcal{C}$ be a vertex cover. Then $|\mathcal{M}| \leq |\mathcal{C}|$.*

10 The approximation algorithm for vertex cover

A matching $\mathcal{M} \subseteq E$ is a *maximal* matching if for every edge $e \in E - \mathcal{M}$, $\mathcal{M} \cup \{e\}$ is no longer a matching.

The algorithm

VertexCoverApprox($G = (V, E)$)

1. Compute a maximal matching $\mathcal{M}$ of G:
   - Set $F = E$, $\mathcal{M} = \emptyset$
   - While $|F| > 0$
     - Add any edge $e \in F$ to $\mathcal{M}$
     - For each remaining $f \in F$, if $|e \cap f| > 0$, remove f from F
2. Let S be the set of nodes adjacent to an edge in $\mathcal{M}$
3. Return S

Theorem 10.1. *Let C^* be the minimum sized vertex cover of G. The algorithm* VertexCoverApprox *runs in polynomial time in terms of the size of G and outputs a vertex cover $S \subseteq V$ satisfying $|S| \leq 2C^*$. Thus, this is a 2-approximation algorithm for vertex cover.*